

MICOSOL 25% SOLUTION

Description and Composition

MICOSOL 25% SOLUTION is a yellow to amber solution. Each ml contains 250mg Tilmicosin as its active ingredient.

Pharmacodynamics

Tilmicosin is a mainly bactericidal semi-synthetic antibiotic of the macrolide group. It is believed to affect bacterial protein synthesis.

Tilmicosin has a wide spectrum of activity against Gram-positive organisms and is particularly active against *Pasteurella*, *Actinobacillus* (*Haemophilus*) and *Mycoplasma* organisms of bovine, porcine and avian origin. Tilmicosin has some activity against certain Gram-negative micro-organisms. Cross resistance between tilmicosin and other macrolide antibiotics has been observed. Macrolides inhibit protein synthesis by reversibly binding to the 50S ribosomal subunit. Bacterial growth is inhibited by induction of the separation of peptidyl transfer RNA from the ribosome during the elongation phase.

Ribosomal methylase, encoded by the *erm* gene, can precipitate resistance to macrolides by alteration of the ribosomal binding site. The gene that encodes for an efflux mechanism, *mef*, also brings about a moderate degree of resistance.

Resistance is also brought about by an efflux pump that actively rids the cells of the macrolide. This efflux pump is chromosomally mediated by genes referred to as *acrAB* genes. Resistance of *Pseudomonas* species and other Gram negative bacteria, enterococci and staphylococci may be precipitated by chromosomally controlled alteration of permeability or uptake of the drug.

Pharmacokinetics

When administered orally to chickens, turkeys and pigs with drinking water and to calves with milk replacer, tilmicosin is absorbed and moves rapidly out of the serum into areas of low pH. This results in very low serum concentrations, but detectable levels of tilmicosin are found in lung tissue as early as 6 hours after starting the treatment. In chicken or turkeys, tilmicosin is also detected in pooled air sac tissue as early as 6 hours after starting the treatment. It is also known that tilmicosin is concentrated in alveolar macrophages of swine. When administered orally to calves tilmicosin is detected in lungs after 6 hours and remains at the therapeutic level up to 60 hours from the last dose.

Indication

Pigs: For the treatment and prevention of respiratory infections associated with *Mycoplasma hyopneumoniae*, *Pasteurella multocida* and *Actinobacillus pleuropneumoniae* when the disease has been diagnosed at the herd level.

Chickens: For the treatment and prevention of respiratory infections in poultry flocks associated with *Mycoplasma gallisepticum* and *Mycoplasma synoviae* when the disease has been diagnosed at the herd level.

Turkeys: For the treatment and prevention of respiratory infections in turkey flocks associated with *Mycoplasma gallisepticum* and *Mycoplasma synoviae* when the disease has been diagnosed at the herd level.

Calves: For the treatment and prevention of respiratory infections associated with *Mannheimia haemolytica*, *P. multocida*, *Mycoplasma bovis* and *M. dispar* when the disease has been diagnosed at the herd level.

Recommended Dose

For oral administration via the drinking water or milk replacer.

Pigs: 15mg to 20mg tilmicosin per kg bw for 5 days, i.e. 6ml to 8ml of product for 100kg bw corresponding to 80ml of product per 100 litres of drinking water for 5 days.

Chickens: 15mg to 20mg tilmicosin per kg bw for 3 days, i.e. 6ml to 8ml of product for 100kg bw corresponding to 30ml of product per 100 litres of drinking water for 3 days.

Turkeys: 10mg to 27mg tilmicosin per kg bw for 3 days, i.e. 4ml to 11ml of product for 100kg bw corresponding to 30ml of product per 100 litres of drinking water for 3 days.

Calves: 12.5mg tilmicosin per kg bw two times per day for 3 to 5 days, i.e. 1ml of product for 20kg bw two times per day for 3 to 5 days.

Route of Administration

For oral administration.

Contraindications

Do not use in case of hypersensitivity to the active substance or in cases of known resistance to tilmicosin. Do not use in horses.

Warning and Precautions

Special warnings for each target species:

Tilmicosin should not be administered by injection to pigs. The product contains disodium edetate. The uptake of medicated water can be altered as a consequence of illness. If the uptake is insufficient alternative treatment may be required.

Special precautions for use

Special precautions for use in animals:

Inappropriate use of the product may increase the prevalence of bacteria resistant to tilmicosin and may decrease the effectiveness of treatment with tilmicosin related substances. Use of the product should be based on susceptibility testing and take into account official and local antimicrobial policies.

Do not allow horses or other equines access to drinking water containing tilmicosin.

Special precautions to be taken by the person administering the veterinary medicinal product to animals

People with known hypersensitivity to tilmicosin should avoid contact with the product. The veterinary medicinal product may cause irritation or sensitisation by skin contact.

Avoid skin and ocular contact. Wear protective gloves and protective clothes when handling the veterinary medicinal product.

In case of contact with skin or eyes, rinse abundantly with fresh water. If irritation persists and in case of incidental ingestion, seek immediately medical advice or call a poison center (dangers linked to disturbances in cardiac conduction).

Wash hands after use.

Interactions with Other Medicaments

Cross resistance between tilmicosin and other macrolide antibiotics and lincosamides has been observed. Tilmicosin may lessen the antibacterial activity of β -lactam antibiotics.

Do not use simultaneously with bacteriostatic antimicrobial agents.

Pregnancy and Lactation

The safety of the product has not been established during pregnancy and lactation. Use only in accordance with risk/benefit assessment by the responsible veterinarian.

Side Effects

None known.

Symptoms and Treatment of Overdose

Pigs drink less water when a dose of 300 to 400mg/liter (1.5 to 2 times the recommended dose) is administered. Although this will result in less intake of tilmicosin, it might lead to dehydration of the animals. Replace with untreated water when needed.

No symptoms were seen in poultry treated at 375mg/liter during 5 days. A dose of 75mg/liter during 10 days resulted in less consistent faeces.

No symptoms of overdose were noticed in turkeys treated at 375mg/liter of drinking water during 3 days. No symptoms were noticed at 75mg/liter during 6 days.

Except for a slight decrease in milk intake, no symptoms of overdose were seen in calves treated at 5 times the recommended dose or during twice the recommended treatment period.

Storage Condition

Store below 30°C.

Protect from direct sunlight.

Shelf life

3 years.

To be used within 24 hours after dilution.

After first opening the container, any product remaining after 24 hours should be discarded.

Withdrawal Periods

Meat and offal:

Pigs: 14 days

Calves: 42 days

Chickens: 12 days

Turkeys: 19 days

Eggs: Not authorised for use in birds producing eggs for human consumption.

Packing

100ml, 200ml, 240ml, 500ml, 1L

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	Manufacturer & Product Registration Holder
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